

risa: Basic functions

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The *risa* package

The **risa** package, which stands for RiSk Assessment, was designed to automate the application of the Habitat Risk Assessment (HRA) model, which is part of the *InVEST* (Integrated Valuation of Ecosystem Services and Tradeoffs) toolkit developed by the Natural Capital Project (Arkema et al., 2014; Sharp et al., 2020).

There are a entire suite of functions that were projected to automate the steps to apply the *ByRA* (Bycatch Risk Assessment) toolkit (Hines et al., 2020; Wilhelm et al., 2022). *ByRA* uses HRA methodology to quantify the spatial overlap between fisheries and marine megafauna and has been applied to assess bycatch risk for marine mammals.

Specifically, **risa** automates the generation of Kernel Density Maps of species occurrences and stressor intensities from either data frames or shapefiles, the creation of overlap hotspot maps, and the execution of the full HRA model for one or multiple species and stressors. Users can select among multiplicative and Euclidean formulations of risk, apply optional decay functions to account for stressor influence buffers, and generate detailed summary statistics. In addition, **risa** provides flexibility to customize model settings, including the number of risk classes, buffer sizes for kernel density estimates and stressor zones, and the choice of coordinate system (metric or decimal).

Installation

Since **risa** is still in its *alpha* version, it cannot be installed from R's CRAN repository. Thus, **risa** must be installed from a GitHub repository, which can be assessed via the link:

<https://github.com/GusmaoJB/risa/>

The easiest way to install **risa** is using the `install_github()` function in the package **devtools**. If you don't have it, install **devtools** and via `install.packages("devtools")` and then run the code below to install **risa**.

```
# Load devtools
library(devtools)

# install the risa package
devtools::install_github("GusmaoJB/risa")

# Load the risa package
library(risa)
```

Performing ByRA

Since we depend on functions in the packages *terra* and *sf* to load and manipulate rasters and shapefiles, we will load these packages to perform our examples. First, let's keep our work files (rasters, shapefiles, dataframes) in a *data* folder inside our working directory. In this first example, let's suppose we have species and stressor distributions as occurrence points in CSV files:

```
# Load important packages
library(sf)
library(terra)

# Loading species and stressor data
spp_df <- read.csv("data/species_distr.csv")
str_df <- read.csv("data/stressor_distr.csv")
```

Then we have a dataframe describing the occurrences of different species:

```
# Check the first rows
head(spp_df)

##           long           lat species group_size
## 1 -4.4417028 -0.30863054 Dolphin             1
## 2  4.7315084 -0.27739997 Dolphin             1
## 3 -2.8702334 -0.14875624 Dolphin             2
## 4 -2.7600157 -0.08447276 Dolphin             3
## 5 -5.9929263 -0.12344416 Dolphin             4
## 6 -0.8168881 -0.24008144 Dolphin             7
```

And a data.frame with similar structure for stressors:

```
# Check the first rows
head(str_df)

##           long           lat stressor
## 1 -0.6628271 -3.4868198  Gillnet
## 2  4.4068230  0.2046171  Gillnet
## 3  1.3602443 -3.2101936  Gillnet
## 4  1.4402246 -6.5176442  Gillnet
## 5  2.2189534 -7.0911457  Gillnet
## 6  1.0967222  2.3932125  Gillnet
```

All we need now is to load the criteria table. This table compiles the key information required for the analysis, including exposure ratings of each species (or habitat) to specific stressors, consequence ratings of those interactions, and the corresponding weights used in risk estimation and data quality assessment.

Since we are working within the **ByRA** framework, we will use the criteria table specifically designed for this purpose, following the approach described in Hines et al. (2020).

```
# Load criteria data
criteria <- read.csv("data/criteria.csv")

# Check the first rows
head(criteria)

##           HABITAT.NAME Dolphin X      X.1 Turtle X.2      X.3
## 1 HABITAT RESILIENCE ATTRIBUTES RATING DQ WEIGHT RATING DQ WEIGHT
## 2           age of maturity          3 3      2      3 3      2
## 3      reproductive strategy          3 3      3      1 3      3
## 4      population connectivity          2 3      2      1 3      2
## 5          local species status          2 3      2      3 3      2
## 6
##      CRITERIA.TYPE
## 1              E/C
## 2              C
## 3              C
```

```
## 4          C
## 5          C
## 6
##
## 1
## 2
## 3
## 4 (3) negligible movement/exchange between the focal regional population and other populations (a DP
## 5
## 6
```

Now we already have everything we need to run our first ByRA!

Let's start with a very simple example, where we just want a quick assessment of bycatch risk for two megafaunal species exposed to two different fishing gears. Since this is a first try, we will keep things straightforward and rely on the default behavior of the `quick_byra()` function.

This means we will not worry (yet) about the details behind how kernel density estimates for species and stressors are constructed, or how the HRA model is calculated. The default settings handle all of that for us, using conservative assumptions and standard data preparation protocols typically recommended for ByRA analyses.

We can always refine and tweak both the data prep and the HRA model later—so for now, let's start simple and see what we get.

```
byra1 <- quick_byra(spp_df, str_df, criteria)
```

And that is it! All the results we want, including the maps and statistics of risk estimates are stored in `byra1`. If we check it out, we will find this:

```
names(byra1)
```

```
## [1] "Dolphin"          "Turtle"           "ecosys_risk_raw"
## [4] "ecosys_risk_classified" "summary_stats"    "kde_maps"
## [7] "area_of_interest"
```

Which means:

- **Dolphin and Turtle:** Partial results (risk estimates) for each species (can be just one or more!). This is a list containing maps of the Exposure and Consequence components of the risk assessment, as well as overall risk estimates for each individual stressor and their combined effects. It also includes a table with summary statistics for these estimates.
- **ecosys_risk_raw:** A `SpatRaster` map of the raw ecosystem risk estimates, representing the combined risk across all species considered in the assessment (Dolphin and Turtle).
- **ecosys_risk_classified:** A `SpatRaster` map of reclassified ecosystem risk estimates. ByRA typically uses four risk categories: 0 = None, 1 = Low, 2 = Medium, and 3 = High.
- **summary_stats:** A `data.frame` with summary statistics of risk estimates at both the ecosystem and species levels. It includes results for individual and combined stressor effects, as well as minimum, maximum, and mean values of Exposure, Consequence, and risk estimates, in addition to the relative area of each risk category.
- **kde_maps:** A list containing all kernel density estimate maps for each species and stressor, as well as maps of overlap for each species–stressor pair.
- **area_of_interest:** An `sf` vector object representing the polygon that defines the area of interest for the ByRA analysis. If not provided by the user, it is estimated by default based on the spatial distribution of the stressors.

First, let's check the summary stats!

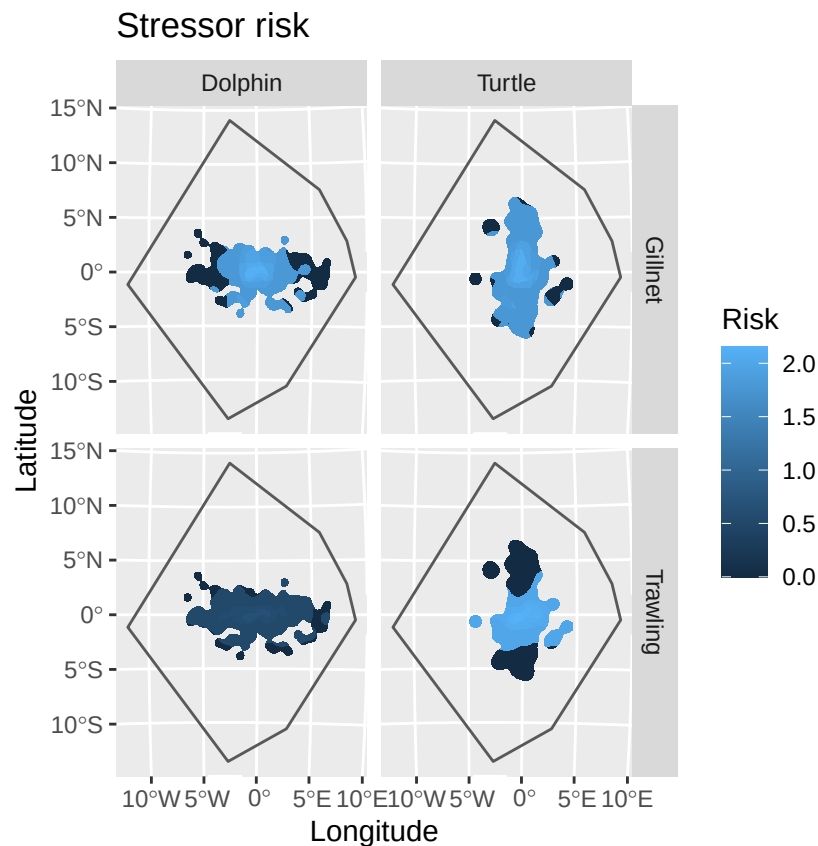
```
byra1$summary_stats
```

```
##      SPECIES      STRESSOR E_min  E_max  E_mean C_min  C_max
## 3 ECOSYSTEM (FROM ALL STRESSORS) NA      NA      NA      NA      NA
## 1 Dolphin (FROM ALL STRESSORS)  0 2.470588 1.352153  0 2.529412
## 2 Dolphin      Gillnet      0 2.470588 1.436176  0 2.529412
## 31 Dolphin      Trawling      0 1.655172 1.268130  0 1.050000
## 4 Turtle (FROM ALL STRESSORS)  0 2.647059 1.619797  0 2.750000
## 5 Turtle      Gillnet      0 2.647059 2.096140  0 2.277778
## 6 Turtle      Trawling      0 2.250000 1.143454  0 2.750000
##      C_mean R_min  R_max  R_mean  R.high R.medium  R.low  R.None
## 3      NA      0 3.9206224 1.8985369  0.283376 46.31083 42.52543 10.880369
## 1 1.1490163  0 2.1217281 0.8076362  0.000000 58.46793 30.01557 11.516508
## 2 1.4961807  0 2.1217281 1.1752274 13.128365 50.46702  0.00000 36.404618
## 31 0.8018519  0 0.6570775 0.4400450  0.000000  0.00000 82.82091 17.179088
## 4 1.6170188  0 2.1505813 1.3196055 26.526288 29.27096 38.86571  5.337041
## 5 1.8046855  0 2.0845908 1.5569531 16.530923 69.40803  0.00000 14.061052
## 6 1.4293521  0 2.1505813 1.0822580 53.463117  0.00000  0.00000 46.536883
```

And now, let's plot the main results using the *risaplot()* function.

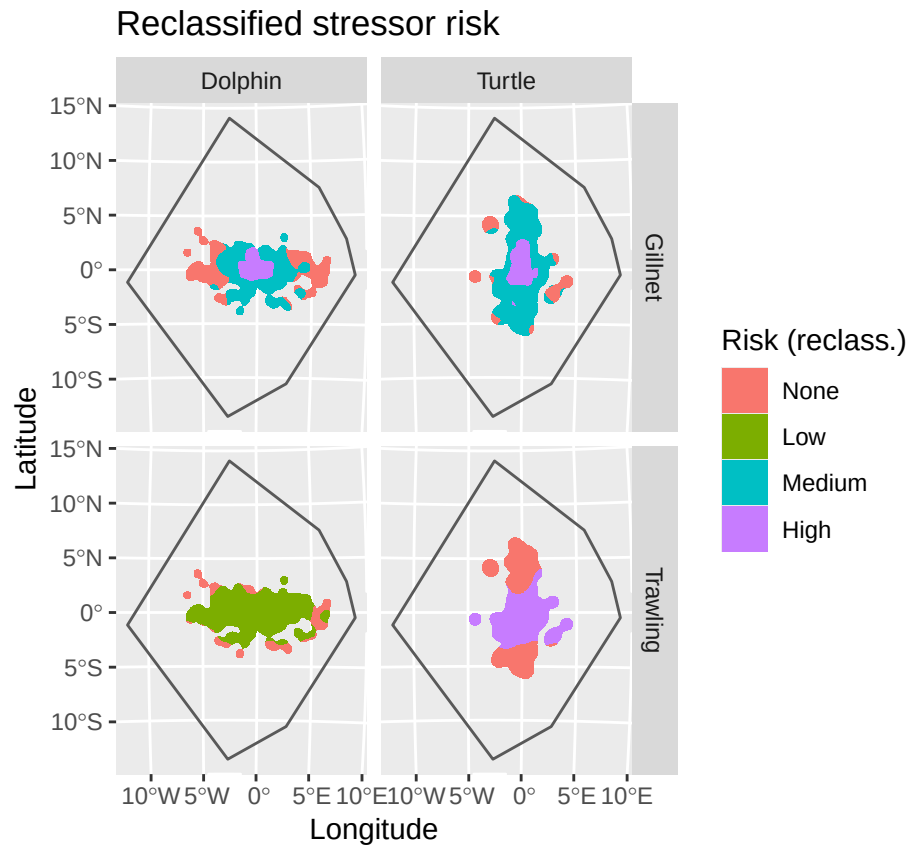
```
risaplot(byra1)
```

```
## [[1]]
```



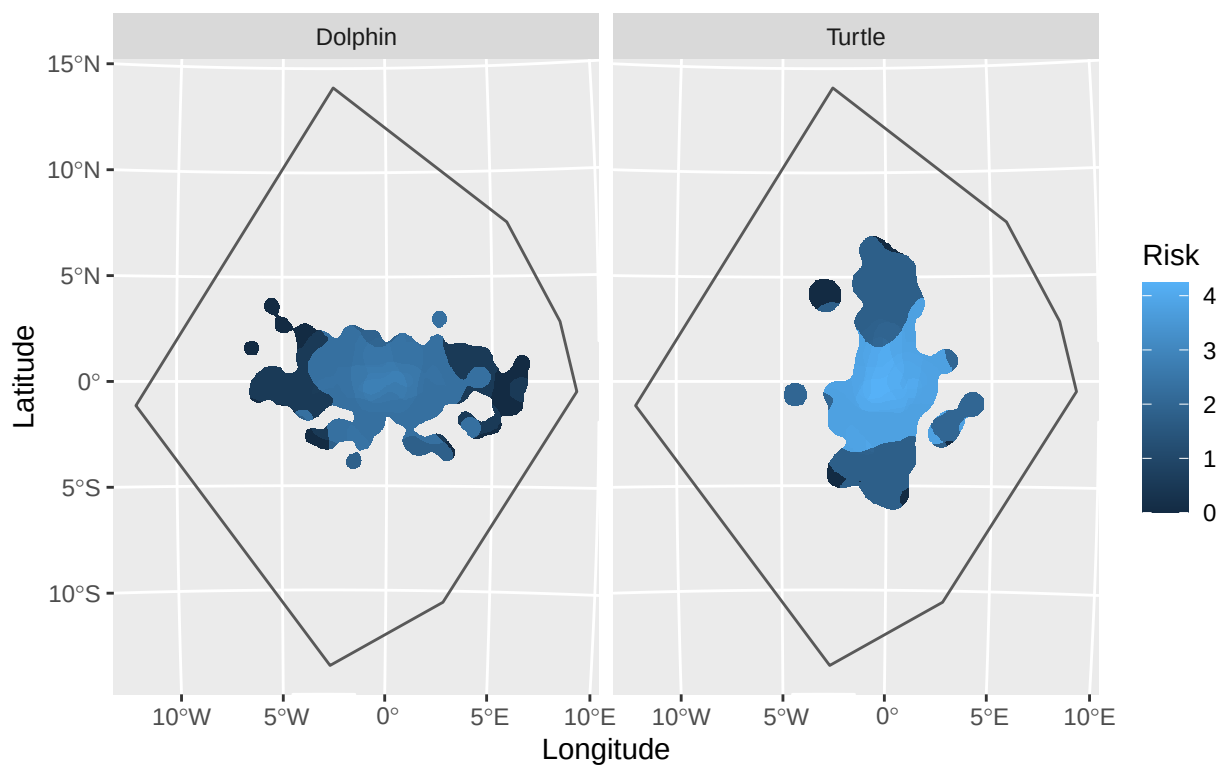
```
##
```

```
## [[2]]
```



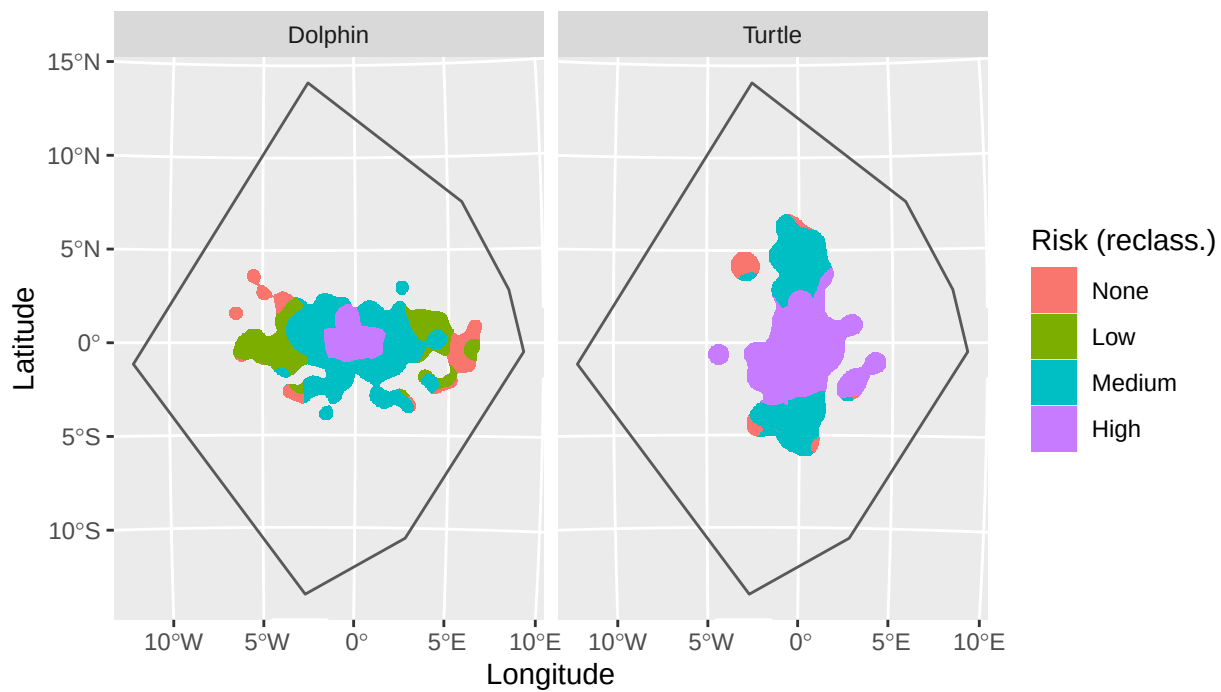
```
##
## [[3]]
```

Total (combined stress.) risk

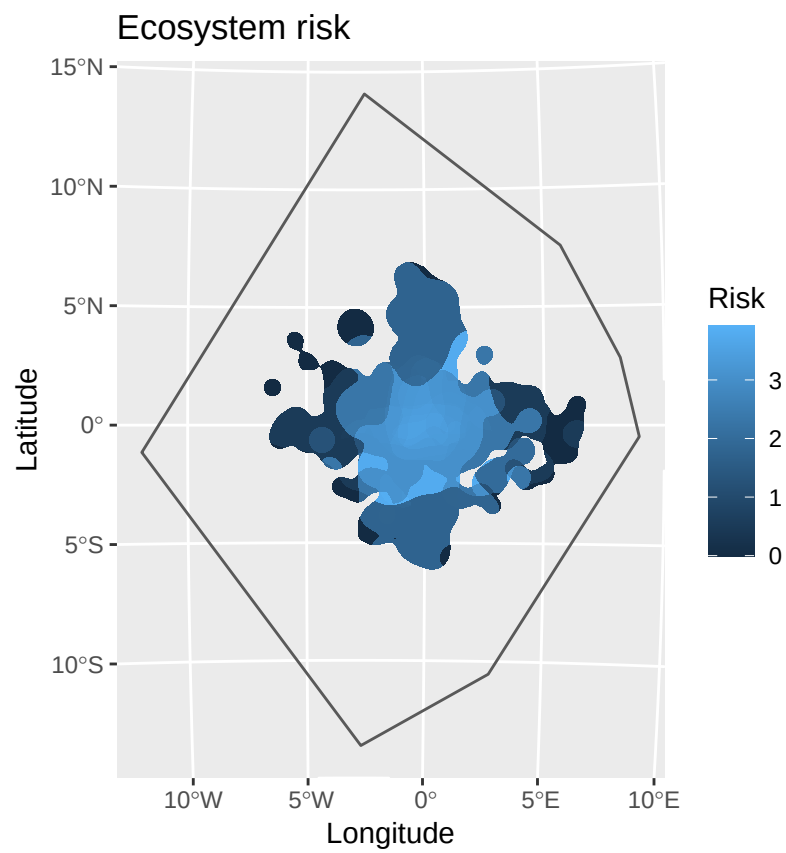


```
##
## [[4]]
```

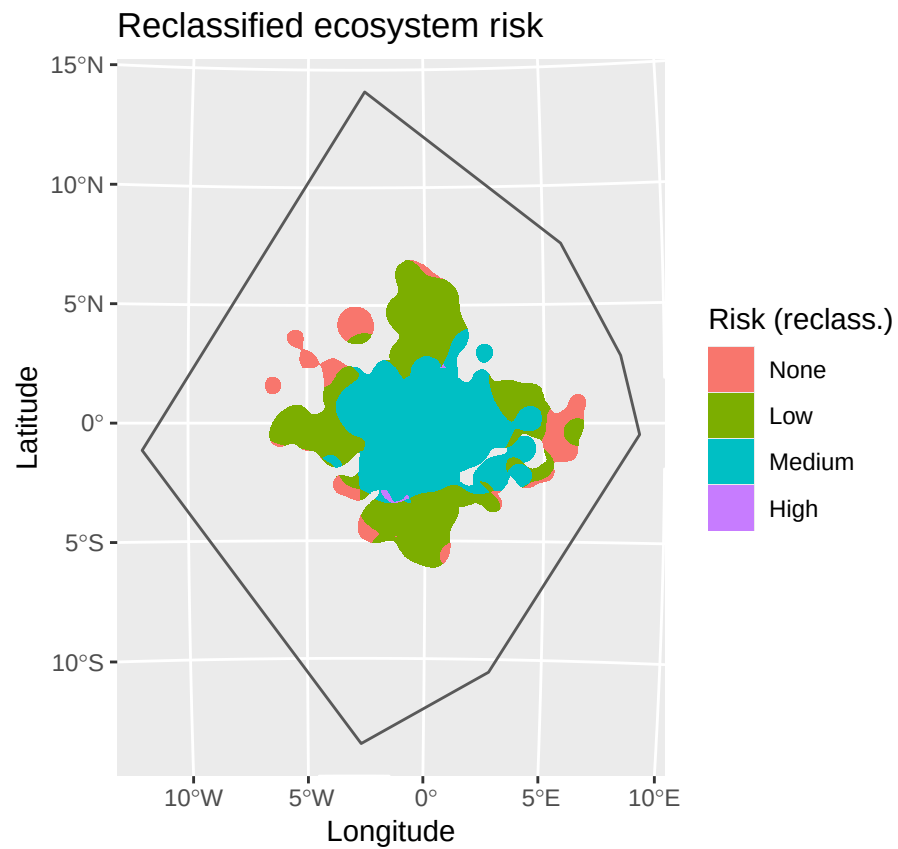
Reclassified total (combined stress.) risk



```
##
## [[5]]
```



```
##  
## [[6]]
```

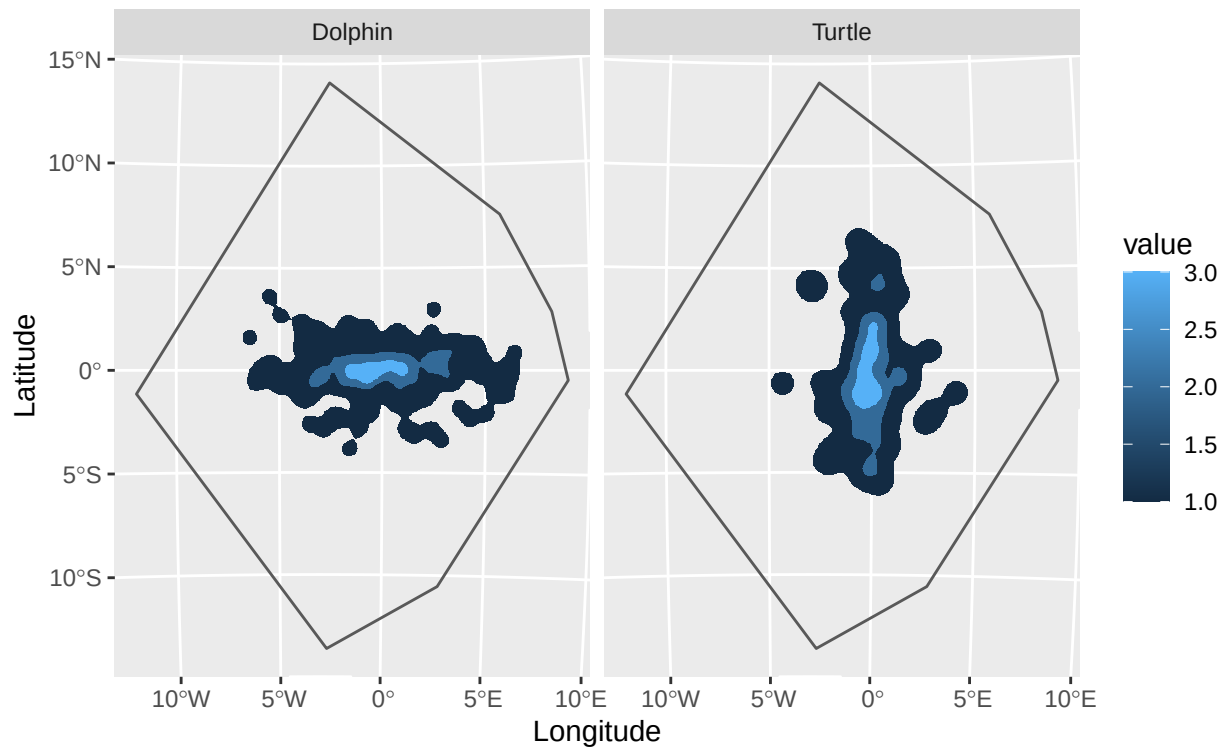


We can also plot the KDE maps and overlaps using `risaplot()` on the `byra1$kde_maps` object.

```
risaplot(byra1$kde_maps)
```

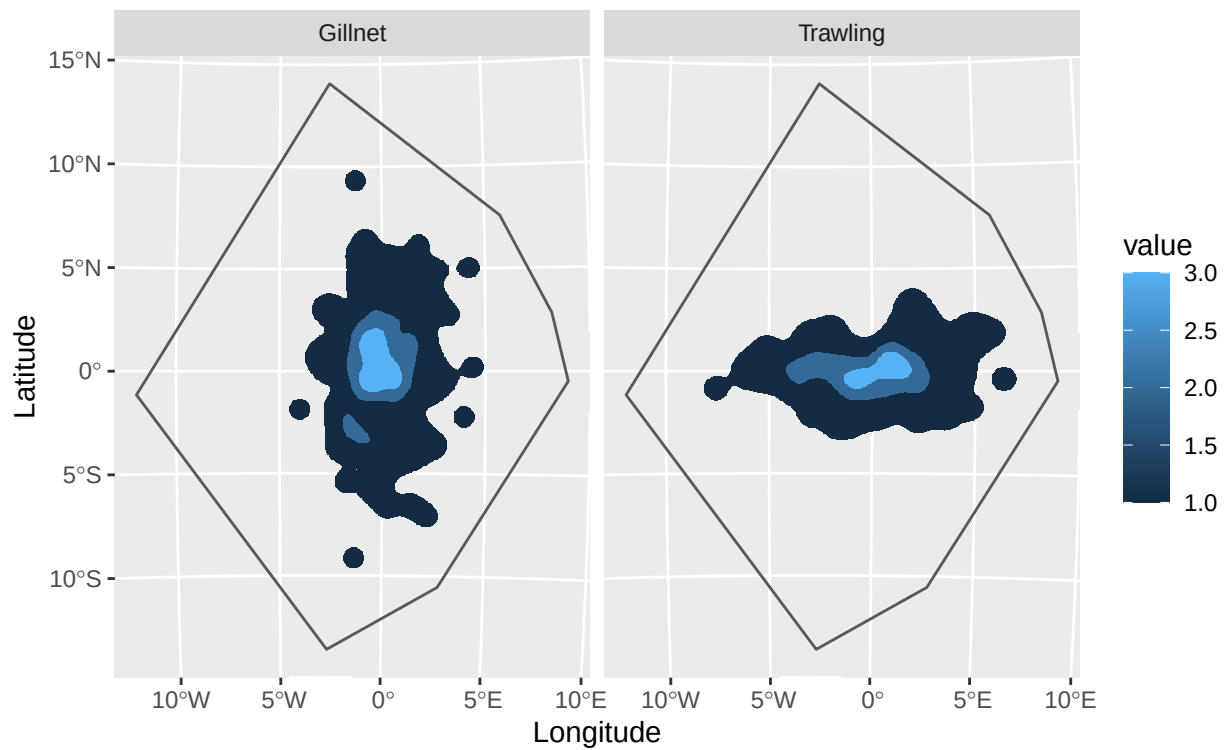
```
## [[1]]
```


Species KDE



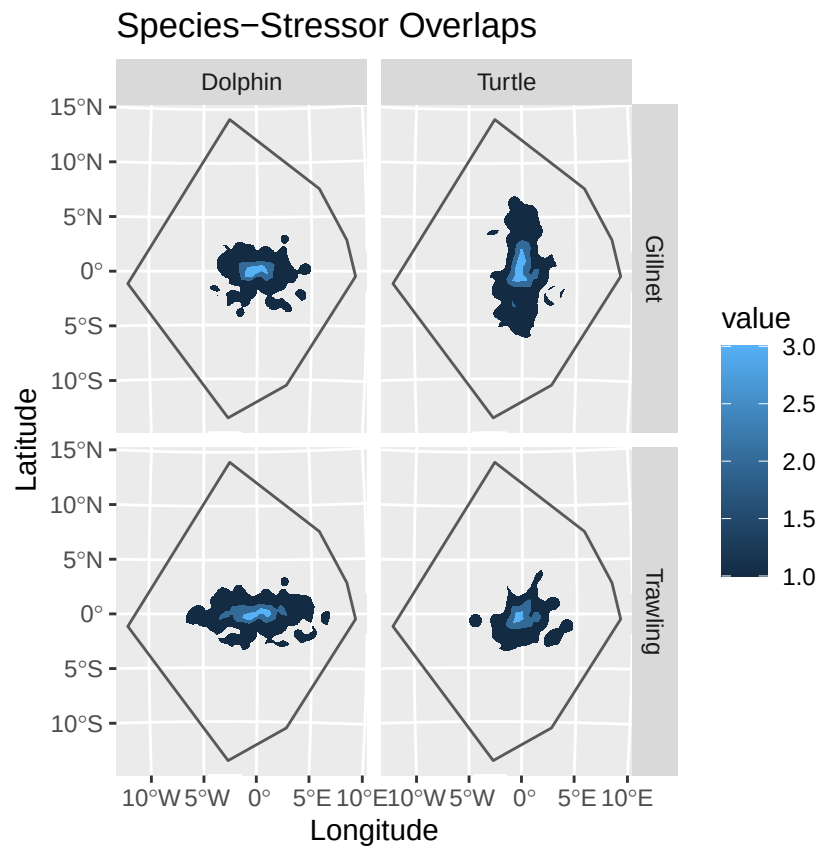
```
##  
## [[2]]
```

Stressor KDE



```
##
```

```
## [[3]]
```



References

Hines, EM, Johnson, A., Ponnampalam, L., Peter, C., Junchompoo, C., Vu, L., Thien, H., Caillat, M., and Verutes, G. (2020) Getting to the Bottom of Bycatch: A GIS-based toolbox to assess the risk of marine mammal bycatch. *Endangered Species Research* 42:37-57 DOI: <https://doi.org/10.3354/esr01037>